

## 1 Gimbal Lock (ie. Euler to Quaternion Motivation)

1.  $\alpha = \text{Roll}$   $\beta = \text{Pitch}$   $\gamma = \text{Yaw}$

$$R = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}$$

$\beta = -\pi/2$

$$R = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}$$

$$R = \begin{bmatrix} 0 & -\sin \gamma & -\cos \gamma \\ 0 & \cos \gamma & -\sin \gamma \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}$$

$\cos(a+b) = \cos a \cos b - \sin a \sin b$   
 $\sin(a+b) = \sin a \cos b + \cos a \sin b$

$$= \begin{bmatrix} 0 & -\sin \gamma \cos \alpha - \cos \gamma \sin \alpha & \sin \gamma \sin \alpha - \cos \gamma \cos \alpha \\ 0 & \cos \gamma \cos \alpha - \sin \gamma \sin \alpha & -\cos \gamma \sin \alpha - \sin \gamma \cos \alpha \\ 1 & 0 & 0 \end{bmatrix}$$

when  $\beta = -\pi/2 = 90^\circ$

$$R_{\text{gimbal}} = \begin{bmatrix} 0 & -\sin(\alpha+\gamma) & -\cos(\alpha+\gamma) \\ 0 & \cos(\alpha+\gamma) & -\sin(\alpha+\gamma) \\ 1 & 0 & 0 \end{bmatrix}$$

2.  $P = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$   $P' = R P = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$  is there an  $\alpha, \gamma$  where this is possible?

$$R = \begin{bmatrix} x & 1 & x \\ x & 0 & x \\ x & 0 & x \end{bmatrix}$$

if  $-\sin(\alpha+\gamma) = 1 \rightarrow \alpha+\gamma = -\pi/2$   
if  $\cos(\alpha+\gamma) = 0 \rightarrow \alpha+\gamma = \pi/2, -\pi/2$

The rotation is possible for  $\alpha, \gamma$  when  $\alpha + \gamma = -\frac{\pi}{2}$ . In this pitch position, the airplane's roll axis is aligned with the yaw axis, so changes in yaw and roll cause the same rotation. We need the plane to rotate  $90^\circ$  about the yaw (Z) axis, and there are infinite  $\alpha, \gamma$  pairs that sum to  $90^\circ$

3. Yes. When the plane experiences this gimbal lock, both yaw and roll cause rotations around the Z-axis. This means that for any rotation, the tip of the wing ( $p'$ ) will be on the XY circle.
4. No. Because yaw and roll are effectively the same dimension, the plane can only rotate about the Z-axis, and not the X-axis. This means that the wing will not be able to be moved to every point on the YZ circle.

These questions illustrate that three values is not enough to represent the rotations because it is possible to get stuck in this gimbal lock scenario where you lose one rotational degree of freedom. A pilot would not be able to safely recover if the plane pitched too much

## 2 Estimating Height with the Time of Flight Sensor

1. When using python libraries to read from the ToF sensor, the measured range is output in millimeters. We need to divide by 1000 to convert to meters, as required by ROS.

## 3 Position Estimation with Affine Transforms

1.  $x_1 = x_0 + E[0][2], y_1 = y_0 + E[1][2], \phi_1 = \phi_0 + \arctan2(E[1][0], E[0][0])$

Because the drone starts at the origin and is initially aligned with the axes of the global coordinate system  $x_0, y_0, \phi_0 = 0$

## 4 Estimating Velocity by Summing Optical Flow Vectors

1.  $\dot{x} = 9ab, \dot{y} = 8ab$

\*Assuming that b contains the information necessary to normalize by the number of optical flow vectors, which may vary in between images.